

Impact Statement Card

AI Saturdays Lagos Cohort 10 | Healthcare SLM Fine-Tuning Safety & Impact Analysis

1. POSITIVE HEALTHCARE IMPACT

Democratized Medical Knowledge

Enables accessible, instant educational information regarding diabetes management, reducing the strain on primary health systems for basic inquiries.

Efficient Low-Resource Compute

By leveraging 4-bit QLoRA on a 2B model, the system can be deployed on lightweight, low-cost edge hardware or single consumer GPUs.

2. RISKS & SAFETY GUARDRAILS

Hallucination Risk

Risk: Potential to generate incorrect dosage or diagnostic advice.

Mitigation: Prepending clinical system prompts and embedding explicit disclaimers in all model outputs.

Automation Bias

Risk: Users trusting assistant output over professional medical advice.

Mitigation: Designing prompt templates that mandate emergency escalation advisories for critical symptoms.

3. ENVIRONMENTAL & COMPUTE EFFICIENCY

Fine-tuning Gemma-2B required only **78 execution steps** using 4-bit quantization, consuming under 10 GB of VRAM. This low computational footprint drastically reduces energy consumption compared to full parameter fine-tuning of larger models (e.g., 70B parameters).

4. LONG-TERM RESPONSIBLE DEPLOYMENT ROADMAP

- **Human-in-the-Loop Validation:** Future iterations will incorporate expert clinician review for safety scoring.
- **Out-of-Domain Detection:** Implementation of guardrails to refuse responses to non-diabetes or complex surgical queries.
- **Open-Source Transparency:** Releasing adapters and preprocessing scripts publicly to foster peer scrutiny and reproducible AI research.